Knowledge Visualization in Qualitative Methods – or how can I see what I say?

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Abstract: **In several research fields, qualitative interviewing (i.e., face-to-face individual interviews and focus groups) is an essential approach to gather high-quality, and deep data about a wide range of subjective experiences. Although being a well-established research method, qualitative interviewing is fraught with various shortcomings such as difficulties in initiating the interview process, in eliciting tacit knowledge and in avoiding information redundancies. In this paper, we propose the use of visual tools (e.g., visualization software) as a support to facilitate the interview process and to gather sound qualitative data. Visual representations provide structure, context and explicitness, and thus foster knowledge elicitation and sharing in interviews. Our contribution is twofold: First, we provide a methodological extension of qualitative interviewing. Second, we suggest a practical approach to use visuals in interviews, presenting concrete examples from the research field. We conclude our paper by discussing the limitations of our approach, and by suggesting directions for future research on visualization-based interviewing.**

Keywords - Qualitative interviewing; one-to-one interviews; focus groups; projective techniques; visual aids; visual facilitation.

1. Introduction

Qualitative interviewing is a valuable research method for gathering data on people's attitudes, opinions, and life experiences [1]. The aim of qualitative interviews is thus to address questions of *how* and *why*, not *how much* or *how many* [2]. Usually, the researcher lets data emerge from the interviewees' accounts and inductively reconstructs clusters of meaning and action [3]. Qualitative interviewing can take a wide variety of forms, being "not a research method, but a family of research approaches that have only one thing in common – conversation between people in which one person has the role of the researcher" [1, p. 2]. While one-to-one interview is most commonly used, qualitative interviewing can also be conducted in larger groups, as in focus groups [4].

Even though interviewing has been the subject of a large body of literature [5, 6] many practical difficulties in conducting interviews have not yet been solved. Visual aids (e.g., cards, maps, photographs) are often used as *projective techniques* to encourage participants to share their innermost feelings or thoughts [7, 8]. In projective techniques, interviewees are asked to freely express their thoughts, by making associations between the visual stimuli and the concept with which they are being presented [14]. In essence, projective techniques evoke reactions to a familiar stimulus (i.e., an image) in order to aid participants elaborate their perceptions about a novel stimulus (i.e., concept).

In this paper, we propose an approach where visuals are used not only as stimuli to elicit comments from participants, but also as instruments to facilitate the interview process. In this *facilitation approach*, visuals are used not as static objects, but rather are drawn simultaneously by the interviewer and the participants. As explained in the next sections, visual facilitation can bring cognitive and communicative advantages to qualitative interviewing, such as enhancing knowledge sharing [9], focusing attention and fostering common understanding [10, 7]. We structure our paper as follows: First, we focus on qualitative interviewing by paying particular attention to the shortcomings of both one-to-one interviews and focus groups. Subsequently, we review the current literature on projective techniques in the attempt to explain the use of visual objects as stimuli in qualitative interviewing. Finally, we describe the advantages of using visuals as facilitation techniques for qualitative interviewing, while also introducing real examples of visual interviews and visual focus groups. We conclude by outlining the limitations of our approach, and by suggesting directions for future research on the use of visual methods in qualitative research.

2. Qualitative Interviewing: Characteristics, Advantages and Shortcomings

2.1. One-to-One Interviews

One-to-one (or individual) interviews are as old as mankind, and had already been used by the Ancient Egyptians for demographic investigations [11]. For individual interviews to be a research method, one of the participants should act as a researcher, and conduct the interview with the objective to answer a research question. Individual interviews can take a large variety of formats, ranging from structured (or close-ended) to unstructured (or ethnographic): While the first gives little room for variation in the answers [11] the latter is closer to observation and leads to open-ended data [12]. In between these two extremes, semi-structured interviews allow for long and in-depth accounts, while also providing guidance on the interview topic.

There are several advantages of one-to-one, face-to-face interviews: First, qualitative interviewing enables the researcher to gain deep insights into the respondents' perspectives [8]. Second, qualitative interviews are relatively inexpensive and allow collection of very rich data, enabling the researcher to notice and correct the respondents' misunderstandings, to probe vague answers, as well

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as to clarify doubts or concerns. Compared to other research methods, face-to-face interviews allow to monitor the order in which the questions are answered, and to control the context of the interview, thereby avoiding the possible biasing presence of other people [8]. Finally, the interviewing methodology is easily adjustable and can be combined with quantitative methods [3].

2.2. Focus groups

The focus group is an interview technique where the researcher collects data by creating a group interactive discussion [13, 14, 15]. This definition deliberately excludes groups or research methods that have inappropriately been called focus groups: First, focus groups should be distinguished from groups set up for purposes other than data collection (e.g., decision making, product development, project management groups). Second, focus groups should be set apart from procedures that involve multiple participants but do not allow interactive discussions, such as nominal and Delphi groups. Finally, focus groups should be distinguished from methods to collect data in naturally occurring discussions where no one acts as an interviewer [15]. Frey and Fontana [4] further suggest to draw a dividing line between group interviews and focus groups: Group interviews cannot be considered as focus groups if they (i) are conducted in informal settings; (ii) use nondirective interviewing; or (iii) use unstructured question formats. To this definition, marketing scholars [16, 14] add another criterion, suggesting that focus groups consist of structured discussions "among 6 to 10 homogeneous strangers in a formal setting" [15, p. 131] As explained by Hoyle et al. [8, p. 401], "Fewer than 6 makes it less likely that the desired diversity of opinions will be elicited, more than 10 makes it difficult for everyone to express their opinions fully … Lack of familiarity promotes free expressions of ideas and opinions that may not be forthcoming if members feel constrained by what they have said in the past to others in the group". Given the breadth of their possible applications, focus groups are used across a variety of academic fields, such as communication studies, education, and public health. Outside of academia, focus groups are popular in marketing research, where they are used to collect consumer's opinions about products, or to estimate the impact of advertising campaigns [17].

Focus groups are low in costs, produce results relatively quickly, and can increase the sample size of a study. However, the main advantage of focus groups lies in the ability to generate *collective data*, as the interviewer asks participants to build on each other, rather than aggregating individual data in search for general patterns. In the words of Morgan [15]: "What makes the discussion in focus groups more than the sum of separate individual interviews is the fact that the participants both query each other and explain themselves to each other". Moreover, the interaction in focus groups – i.e., the "group effect" – enables the investigation of complex phenomena, that are more difficult to uncover in one-to-interviews. Morgan and Krueger [18] have also underscored that such group interaction provides valuable data on the extent, and the nature of consensus and diversity among participants.

2.3. Shortcomings of Qualitative Interviewing

There are several challenges in qualitative interviewing, such as getting the interviewee to speak freely, and gathering high-quality and valuable data. During the interview process, one of the greatest challenges is managing information, which means eliciting tacit knowledge, and at the same time avoiding information redundancy. Fontana and Frey [11, p. 361] describe the process-related difficulties of qualitative interviewing in a vivid and concrete way: "Asking questions and getting answers is a much harder task than it may seem at first. The spoken or written word has always a residue of ambiguity, no matter how carefully we word the questions and report or code the answers". Besides issues of wording and language choice, the phenomenon of *knowledge asymmetry* often threatens to disrupt the interview flow. In some instances, the knowledge of the interviewee – especially if she is an expert in her field – is much larger than the foreknowledge of the interviewer. This discrepancy leads to misunderstandings which usually are not detected prior to analyzing the data. At this stage, contacting the interviewee for further clarifications or questions may be practically unfeasible. The problem of knowledge asymmetry may occur also in the other direction, when the interviewer – because of her confidence with the interview topic – uses an expert language which is too elaborated for the counterpart to understand. If the interviewee is unable to fully understand the question, she may fear to lose her status when probing the question, and thus may provide inaccurate or vague answers.

Another difficulty lies in building up trust with the interviewee, and gaining access to deep and high quality data. If the interviewer fails to build a personal chemistry with the counterpart, for example by appearing unprofessional, the respondent may hide information. Furthermore, *interviewer effects* may interfere in the rapport between the interviewer and the respondent: The interviewer's expectations or personal characteristics, such as ethnicity, physical appearance and sex, can bias the responses. Since a strong interpersonal rapport is created in face-to-face interviews, respondents often tend to please the interviewer by providing socially desirable, yet invalid answers [8].

In focus groups, the moderator is faced with further challenges, such as encouraging participants to build on each other, avoiding overlaps in turn taking, and preventing individuals from dominating the conversation. In addition, there may be *conformity pressure* operating in a focus group: In this case, the moderator should identify nonverbal signs that indicate that a participant disagrees with what is being said. Besides, it is undeniable that group members affect each other in their answers [15, 19]. Sussman et al. [19] found a *polarization effect* related to group interaction – i.e., the attitudes of participants become more extreme after the group discussion. Moreover, the *mutual exposure* of participants limits the range of topics that can be effectively investigated in focus groups.

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Since group interaction requires self-disclosure, some participants may find certain topics unacceptable for discussion [15].

Given the inherent challenges of qualitative interviews, the skillfulness of the interviewer / moderator has important consequences for the quality of the data collected. In focus groups, the moderation role is particularly difficult, since the facilitator is required to guide a multilateral conversation and at the same time to carefully process what is being said. Finally, a common challenge in both focus groups and interviews is to organize, and analyze the data gathered in order to identify recurrent patterns of meaning [8].

In the next sections, we make an attempt to explain how visual aids may help to overcome the multiple challenges of qualitative interviewing. We start by reviewing the current literature on projective techniques, and outline the advantages and the limitations of using visuals as stimuli in qualitative interviews. We then argue for a paradigm shift in the role of visuals, suggesting that the use of visual facilitation may improve qualitative interviewing, and increase the quality of data collection.

3. Visuals as Stimuli in Qualitative Interviews

Visual aids, such as drawings, pictures and maps, can be used during the qualitative interviewing as projective techniques to elicit comments from the participants [8]. Visual aids remove respondents' natural inhibitions, reveal less conscious and deeply-seated feelings, while also allowing to add affective evaluations to rational evaluations. As illustrated by Anastas [21], visuals jump-start respondents' imaginations and verbal responsiveness: "When asked a question like 'what are your feelings when you use this product?', some respondents become tense and anxious … When asked 'select a picture and tell how it shows your feelings when you use the product', they relax and jump into complex stories". By engaging interviewees, visuals enable better access to the participants' thoughts, energize the interview process and in turn lead to high-quality data [20]. Moreover, visuals act as *boundary objects* [22], i.e. as objects that facilitate the sharing of tacit knowledge, and reduce knowledge barriers between the interviewer and the participants. In reviewing the current literature on projective techniques [14, 23], we have identified four typologies of visual-based stimuli:

- *Association*. In the association technique, photographs are used to stimulate the participants' thinking, and to help them articulate their feelings about the discussion topic. In general, the interviewees are asked to express their impressions about pictures portraying people (personality associations), or situations (situational associations) that are connected to the discussion topic. In alternative, the participants may be invited to indicate which of several photographs in a stack most closely relate to the subject under discussion (forced associations). This technique enables the interviewer to uncover deep insights, as the participants tend to project their feelings of the discussion topic onto the pictures stack.
- *Expression*. In expressive drawing, participants are asked to provide their reaction to a discussion topic by drawing a picture of how they feel about it. At the end of the task, the interviewer asks the respondent to comment on the meaning of her picture, and to elaborate on the relationship with the discussion topic. In focus groups, the moderator can involve the entire group in developing interpretations of each other's drawings, and hence of the discussion topic. Expressive drawing can be very useful to elicit information that otherwise might not be generated with traditional techniques, and at times can also motivate respondents.
- *Completion*. Another type of projective technique used in qualitative interviews is picture completion. In this technique, the participants are presented with an incomplete comic-strip story and are asked to add statements into the empty balloons. Like other projectives, picture completion helps the interviewer delve into the participants' minds, and learn more about their inner feelings.
- *Grouping*. In the grouping technique, participants are asked to collect, and to group a number of images that relate to the topic being discussed. The interviewees – either individually or in group – are given a pile of newspapers, scissors, glue and a blank piece of paper onto they can make a collage of pictures to represent the discussion topic. At the end of the task, the interviewees are asked to comment on the collage, and to explain the rationale behind their image choices. Often used in marketing research, this technique has proven useful to elicit the values, typical users, and purchasing motives associated to new products.

Although providing high-quality and deep data, the visual projectives described above fail to fully address the multiple challenges normally encountered in qualitative interviewing. In the following section, we explain how a paradigm shift from visuals as *projective techniques* towards visuals as *facilitation techniques* may enhance qualitative research, by enabling interviewers to fully exploit the potential of visual language.

4. Visuals as Facilitation Techniques in Qualitative Interviewing

As mentioned above, visuals can be introduced into qualitative interviews not only to elicit comments from participants (i.e., content support), but also to improve the research methodology (i.e., process support). In the facilitation approach, visuals are drawn simultaneously by the researcher and the participants during the qualitative interview. The visual language is thus integrated into the research process, and shifts from being an external stimulus to prompt answers, to becoming an integral component of the research method. As an example, the facilitator in focus groups may use visualization tools such as visual

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templates, sticky walls, and e-moderation software to lead the discussion, and at the same time enable participants to write and/or draw out their thoughts. In particular, visual templates – printed on a large poster support – provide a structured canvas where participants can document the discussion with post-it notes.

In comparison to purely text-based facilitation (e.g., flipchart handwriting), visual facilitation enables greater visibility of discussion threads, while also providing a more engaging experience for participants. Moreover, visual representations play a crucial role in the sense-making of individuals and groups, by bringing the advantage of *representational guidance*. In fact, the notations on a visual template may constrain what is expressed, and at the same time make certain aspects more salient. Thus, visual representations may support qualitative interviewing by providing an overarching structure which organizes information, coordinates the conversation, and highlights key aspects [24]. As suggested by Kuchenmüller and Stifel [25, p. 394], by filling visual templates the individual and / or the group "answers important questions while respecting definite rules".

By drawing attention on central themes, visual facilitation helps both the researcher and the interviewee to keep track of the conversation threads. In focus groups, this leads to the constitution of a *group memory*, which reduces the participants' difficulties in following the conversation, and building on each other's contributions. This advantage is present also in *online interviews* and *focus groups*, where visual techniques enable participants to make more fitting contributions, by providing tangible orientation and signposts during the conversation.

Furthermore, visualization techniques have been proven to facilitate balanced participation of group members, and to reduce the potential for interpersonal conflict. This is the case, since the target of the conflict shifts away from the person, to the concept being mapped onto the visual canvas. The *de-personalization effect* of visuals is particularly beneficial in focus groups, for reducing social pressure and encouraging participants to freely express their dissent with the rest of the group. When the conversation is mediated by visuals, participants are not pointing directly at each other, but rather express their agreement or disagreement towards impersonal objects, such as notes on a visual template [26]. In a study of business meetings, Kuchenmüller and Stifel [25, p. 388] consistently noticed that "it is easier to offend and attack using words than images". At the end of the qualitative interview, showing the filled template to the participants offers the possibility to gather further data, by asking probing questions, delving into important aspects, and commenting on the main findings.

As regards the final stage of data analysis, the use of prefabricated templates facilitates the *aggregation*, and the *comparability of data* collected across multiple interviews or focus groups. Thus, the use of visual templates leads to more reliable results, while also reducing the risk of gathering highly idiosyncratic data that cannot be easily inserted into the broader framework of the research theme.

Based on our literature review, as well as on our experience in qualitative interviewing, we suggest a four-stage method for visual-based research in one-to-one interviews and focus groups:

1. Preparing visual templates: Before the qualitative interview, the researcher should carefully choose visual templates based on the topic under investigation. As an example, a visual template for expectation inquiry may be useful in the initial phase of a focus group, to explore participants' expectations, experiences, perplexities, and questions on the subject at hand. For the selection of visual templates, the extant literature on knowledge visualization may provide a useful reference [27].

2. Filling visual templates: During the qualitative interview, the researcher fills the visual template by prompting answers from the interviewee(s). As regards group interviews, guiding and at the same time visualizing the conversation may be exceptionally difficult, and it is therefore advisable to have two facilitators acting in combination, one to moderate and the other one to visualize on her behalf [25].

3. Wrapping-up with visual templates. At the end of the qualitative interview, the researcher can summarize the main points emerged in the interview process by showing the filled template to the interviewees. By wrapping up the qualitative interview with visual representations, the interviewer may stimulate respondents to further delve into relevant topics, or to clarify misunderstandings in the documentation process.

4. Analyzing visual data. The researcher may look for consistent patterns by integrating the visual data reported on multiple templates. If necessary, she may explore group differences by comparing the visual templates filled out by different categories of interviewees and participants. In a way, the availability of standard templates replicates the rigor of quantitative procedures for data collection, although without trading away the richness and deepness of qualitative data.

In the next section, we present a few examples from our research experience, to better illustrate the use of visual facilitation in one-to-one interviews and focus groups.

5. Examples from the Field

In two large research projects, we have facilitated qualitative interviewing situations with a total of more than 60 participants. In all of these face-to-face interviews (one-to-one and focus groups) we have been using visualizations for three purposes: First, to stimulate respondents and thus to initiate the interview process; Second, to sharpen the quality of data by eliciting deeper information; and third, to moderate the interview process with considerable improvements from a methodological perspective. In the following sections, we will present examples and discuss results from both interview situations.

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5.1. Visualizations in One-to-One Interviews

In the first research project, we have conducted qualitative interviews for the purposes of gathering expert information on the use of software applications for monitoring natural hazards in Switzerland. Challenges at the very beginning of one-to-one interviews, such as establishing rapport, gaining trust, and understanding the language of respondents, could be overcome by using an association stimulus, i.e. a sketch about the interview process (Fig. 1). This sketching was used not only as an 'ice-breaker' at the beginning of the interview, but also as a reference to structure the interview content, and to ensure coverage of all the discussion topics (e.g., problems in software appropriation).

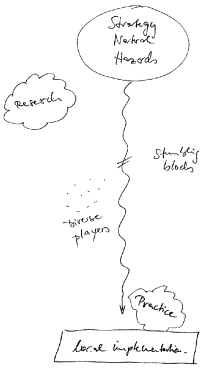

Figure 1: A sketch used as stimuli for expert interviews.

During the interview process another visual tool was used, namely a visual timeline loaded on a graphic facilitation software (Fig. 2). The respondents were asked to fill out the empty template by indicating how they perceived the evolution of their work domain including events, laws, and directives. This completion stimulus addresses the challenge of gathering high-quality data, by literally bringing tacit knowledge, experiences and opinions to the table. All of the participants liked the template completion, and provided positive feedback without being asked. Often, there was a moment of surprise in the beginning, which then turned into 'fun' and an ambition to fully complete the sheet. While filling out the visual, the interviewees kept on talking, what we consider as 'thinking aloud'. This phenomenon is reminiscent of ethnographic interviewing [11].

Based on our experience, we observed three main advantages of using visualization in one-to-one interviews: First, the respondents shared insights, experiences and memories which probably would have been hidden or forgotten without the template. Second, the template use allowed deep insights into the opinions and world views of the respondent and thus generated high-quality data. Third, the visuals enabled us to note side information mentioned incidentally by the interviewee, and in turn to elicit nuances that often lead to highly important and unexpected findings. Last but not least, the interviewer using visuals appeared to be more reliable and was regarded as being professional, objective and clear in her communication.

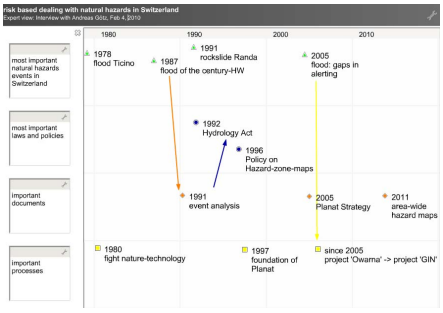

Figure 2: Timeline used as visual facilitation during the interview and for data analysis in retrospect.

5.2. Visualizations in Focus Groups

In a second research project, we have investigated the use of software applications for supporting decision-making in inter-organizational settings. In terms of research design, we have used randomized experiments as a primary method, and introduced focus groups as a follow-up method. In particular, we have used focus groups with the purpose to clarify unexpected experiment findings, while also gathering richer and deeper data on the research topic (see [15, 17] on the use of focus group in combination with other research methods). The experimental subjects served as focus group participants and were asked to reflect on their experience in the experiment.

In conducting the focus group, the facilitator used a visualization software (let's focus) connected to a video projector. For each interview question, the facilitator filled a visual template with contribution from the focus group participants (an example is shown in Fig. 3). In this regard, visual templates were used for real-time documentation of the focus group discussion. The visual facilitation enabled focus group participants to keep track of the discussion, and to build on each other's contributions with greater ease. Furthermore, the use of visual templates ensured consistency and comparability of data collected across different focus group sessions. As regards the data analysis, the availability of standard templates facilitated the comparisons, and aggregation of qualitative data. In this regard, the researchers elaborated an aggregated visual template summarizing the common patterns that emerged across the focus group sessions.

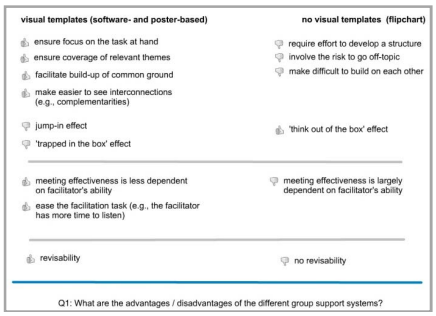

Figure 3: Visual template used for real-time documentation in focus group

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6. Discussions and Conclusions

Although providing considerable advantages, qualitative interviewing is fraught with methodological shortcomings. The major difficulties are situated in initiating the interview process, capturing valuable data, and preventing respondents to go off topics. In this article, we have proposed an interdisciplinary procedure, suggesting to use visual methods in qualitative interviews to obtain deeper insights. The visuals, used both as stimuli and facilitation, work as 'boundary objects' [22] and therefore enhance discussions, debates or consensus during the interview situation. This allows for more awareness of the respondents' perceptions, and thus permits to draw more rigorous conclusions. However, some limitations exist in the use of visuals for qualitative interviews and focus groups, as for example:

- The productivity of the interview is highly dependent on the visual literacy of the interviewer or moderator.
- Given the difficulty of simultaneously conducting and visualizing the interview, it is advisable to have two facilitators or interviewers working together. Nevertheless, the coordination among the two researchers may be difficult, or bias the interview process.
- The respondents may be 'locked' into the visual templates, and neglect discussing relevant issues that are not pointed out on the visual structure. This is the negative side of the representational guidance provided by visual representations [24].
- The visuals may 'cajole' the respondents, and lead to overly positive attitudes towards the interview topic [28].

Moreover, the mentioned strengths of visualization methods in qualitative interviews need to be substantiated with further empirical evidence. In future studies, for example, researchers could assess the added value of visualization-based interviewing through an experimental methodology, i.e. by comparing the quality of data gathered in intervention (visualization-based interviewing) versus control (pure verbal interviewing) groups. Also, our practical methodology for using visualizations in one-to-one interviews and focus groups has to be further tested, evaluated and discussed in academia. Nevertheless, it should be mentioned that our preliminary experiences make it evident that the real-time use of visualization enables people to better see what they say and mean.

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